Multivariate Volatility Modelling

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M2 MBFA

Introduction

Multivariate GARCH
- Motivation
- Definition
- Variance and covariance models
- Variance and correlation models

Estimation methods

Miscellany

Outline

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Financial returns

- Let p_{it} denote the **price** of asset i at date t.
- The (one-period, i.e. from $t-1$ to t) **return** is defined by

$$y_{it} = \log p_{it} - \log p_{i,t-1} = \Delta \log p_{it} \approx \frac{p_{it} - p_{i,t-1}}{p_{i,t-1}}$$

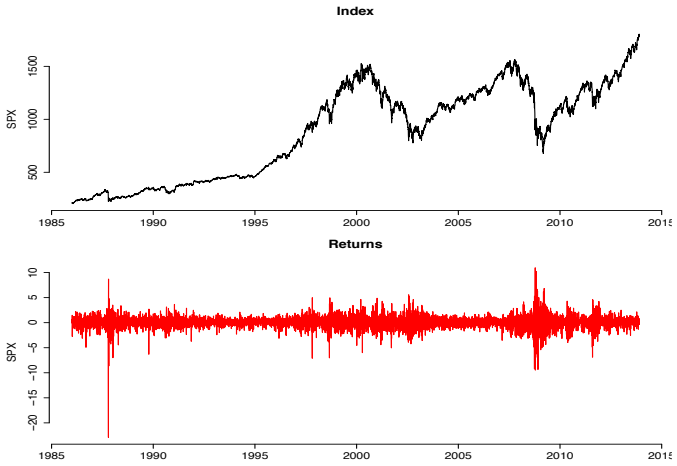
  This may be multiplied by 100 to express the return in percentage.

- The frequency of a return series refers to the length of the time period between t and $t-1$.
  - Low frequency: less than weekly, e.g. **monthly**.
  - High frequency: weekly to **daily**.
  - Ultra high frequency: **intradaily** (less than daily).

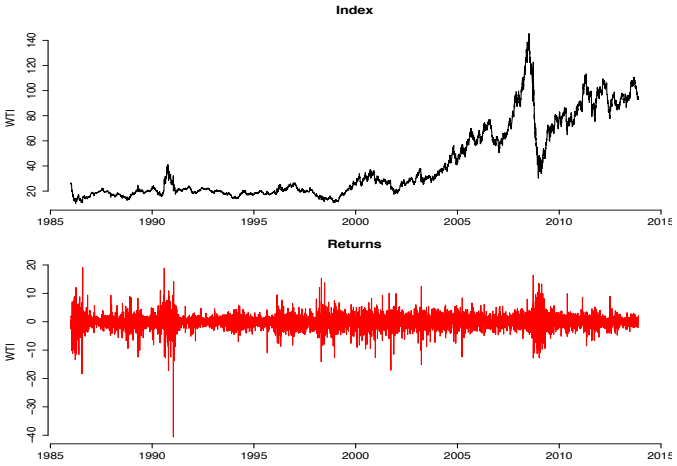
Notations

$$p_t = \begin{pmatrix} p_{1t} \\ p_{2t} \\ \vdots \\ p_{Nt} \end{pmatrix} \qquad y_t = \begin{pmatrix} y_{1t} \\ y_{2t} \\ \vdots \\ y_{Nt} \end{pmatrix} = \Delta \log p_t$$

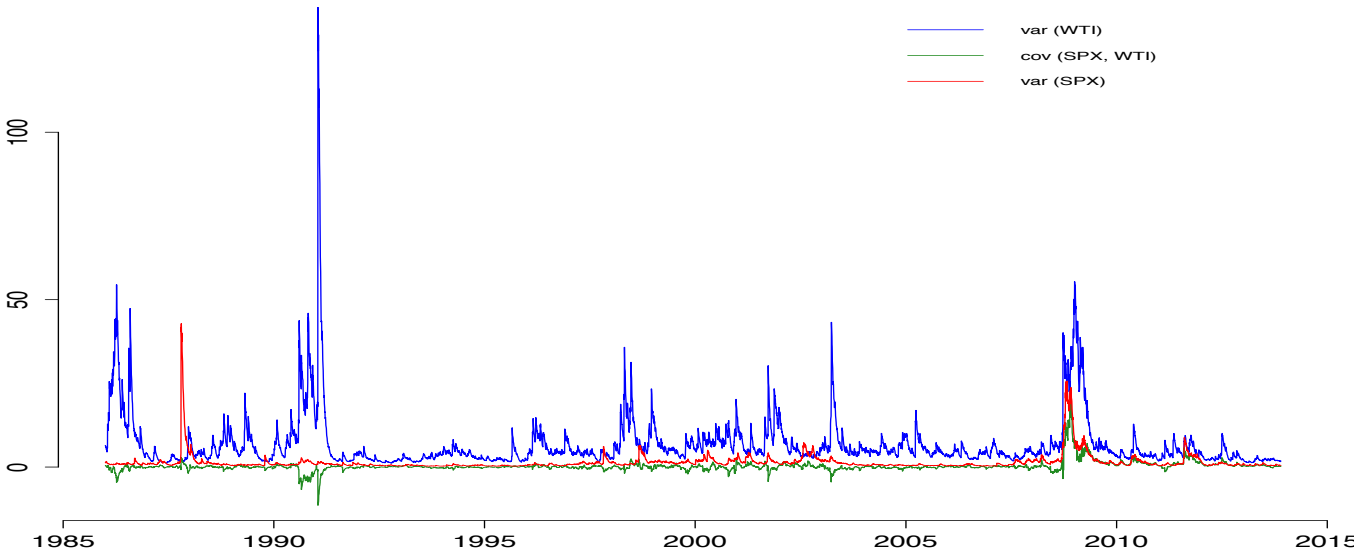
Introduction



(a) S&P500



(b) WTI



(c) Conditional (co)variances

Volatility

- ⋆ The volatility of an asset alludes to the fact that the asset **price** and therefore its associated **return** are **random**. Randomness implies we can **never** be sure of the future value of the asset (or the return).
- ⋆ Technically, the **volatility** of an asset is the standard deviation (or the **variance**) of the return considered as a **random variable**.
- ⋆ Likewise, the **covolatility** between two assets is the **covariance** between the two corresponding returns.
- ⋆ **Returns** and (**co**)**volatilities** are defined relatively to a time period of some length (day, week, year, ...).

Volatility is latent

- ⋆ Volatility is inherently **not observable**.
- ⋆ Like returns, it evolves **randomly** through time. This does not imply that it is not predictable.
- ⋆ There is a double **uncertainty** in financial markets: about returns, and about the volatility of returns.
- ⋆ Volatility can be **indirectly measured**, since returns are observed. A measure can be "model-free" or "model-based".
- ⋆ How to infer about volatility from returns has been and will be the main research theme of financial econometrics for long.

Model-based measures of volatility

- ⋆ RiskMetrics.
- ⋆ Multivariate GARCH models.
- ⋆ Multivariate stochastic volatility models.
- ⋆ Each type of model specifies the volatility matrix as a **parametric function** of past returns and maybe other information. Given the relevant information and an estimate of the parameters, prediction is feasible.
- ⋆ The challenge of ensuring a Positive Definite (PD) variance matrix arises.

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Motivation

- ⋆ Portfolio allocation
- ⋆ Value-at-Risk in a multivariate context
- ⋆ Futures hedging
- ⋆ Volatility transmission
- ⋆ Asset pricing models

Motivation

- ⋆ Multivariate GARCH models: Modelling individual time series separately leaves out information about **comovements** between assets
- ⋆ Large amount of multivariate model handling the **conditional second moments** of (financial) **time series**.

  Surveys: Bauwens, Laurent and Rombouts (2006), and A. Silvennoinen and Teräsvirta (2009).

The general set up

Let the observations $\{y_t\}$ be **multivariate** time series of R^N defined by

$$y_t = \mu_t + \varepsilon_t, \qquad \text{for} \quad t = 1, \ldots, T \tag{1}$$

where the conditional mean $\mu_t = \mathrm{E}(y_t | \mathscr{F}_{t-1}) = 0$ to simplify the discussion, and $\mathscr{F}_t$ is the σ-field generated by all the information until time t.

Consider ε_t an $N \times 1$ disturbance vector with conditional mean

$$\mathrm{E}(\varepsilon_t | \mathscr{F}_{t-1}) = 0$$

and conditional variance

$$\mathrm{E}(\varepsilon_t \varepsilon_t' | \mathscr{F}_{t-1}) = H_t;$$

H_t is the **conditional covariance matrix** defined by $H_t = H_t(\theta)$ where θ is a parameter vector.

We define a **multivariate GARCH** process such that

$$\varepsilon_t = H_t^{1/2} z_t, \tag{2}$$

where $z_t \sim iid(0, I_N)$, and I_N is the $N \times N$ identity matrix.

Remarks

- ⋆ $H_t^{1/2}$ is any $N \times N$ matrix such that H_t is the conditional variance matrix of y_t (e.g. $H_t^{1/2}$ may be obtained by a Cholesky factorization of H_t).
- ⋆ μ_t and H_t depend on **unknown parameters** θ but are otherwise known (parametric model), hence sometimes we write explicitly $\mu_t(\theta)$ and $H_t(\theta)$. For example, μ_t may be a VARMA model.
- ⋆ The "z_t is iid" assumption may be relaxed to "z_t is a martingale difference sequence (MDS) with respect to $\mathscr{F}_{t-1}$", e.g. to show that $\text{Var}_{t-1}(y_t) = H_t$. However, for ML estimation, the iid assumption is relevant.

Three challenges

- ⋆ **State conditions** on the parameters θ such that $H_t > 0, \forall t$, given that $H_0 > 0$
- ⋆ **Avoid too many parameters** (to keep estimation feasible), but maintain enough flexibility in the dynamics of H_t.
- ⋆ **Find the unconditional covariance** $H = \mathrm{Var}(y_t) = \mathrm{E}(H_t)$ and the conditions for the existence of H and for weak stationarity of y_t

For ease of exposition, $H_t = (h_{ijt})$ is a function of one lag of ε_t and one lag of itself, i.e. we review MGARCH(1,1) models.

Typology of MGARCH models

- ⋆ Models that specify conditional variances and covariances as linear functions of past: multivariate generalizations of univariate GARCH(1,1) model

$$h_t = \omega + \alpha \varepsilon_{t-1}^2 + \gamma h_{t-1}$$

  - ⋆ VEC
  - ⋆ BEKK

- ⋆ Models that specify conditional variances and correlations:
  - ⋆ CCC
  - ⋆ DCC

VEC(1,1) Bollerslev, Engle, and Wooldridge (1998)

In the bivariate case,

$$
\begin{aligned}
h_{11t} &= c_1 + a_{11}\varepsilon_{1,t-1}^2 + a_{12}\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{13}\varepsilon_{2,t-1}^2 \\
&\quad + g_{11}h_{11,t-1} + g_{12}h_{21,t-1} + g_{13}h_{22,t-1} \\
h_{21t} &= c_2 + a_{21}\varepsilon_{1,t-1}^2 + a_{22}\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{23}\varepsilon_{2,t-1}^2 \\
&\quad + g_{21}h_{11,t-1} + g_{22}h_{21,t-1} + g_{23}h_{22,t-1} \\
h_{22t} &= c_3 + a_{31}\varepsilon_{1,t-1}^2 + a_{32}\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{33}\varepsilon_{2,t-1}^2 \\
&\quad + g_{31}h_{11,t-1} + g_{32}h_{21,t-1} + g_{33}h_{22,t-1}
\end{aligned}
$$

Matrix notation for bivariate VEC(1,1)

$$\begin{bmatrix} h_{11t} \\ h_{21t} \\ h_{22t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} \varepsilon_{1,t-1}^2 \\ \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{2,t-1}^2 \end{bmatrix}$$
$$+ \begin{bmatrix} g_{11} & g_{12} & g_{13} \\ g_{21} & g_{22} & g_{23} \\ g_{31} & g_{32} & g_{33} \end{bmatrix} \begin{bmatrix} h_{11,t-1} \\ h_{21,t-1} \\ h_{22,t-1} \end{bmatrix}$$

Remark: The number of parameters is of order N^4
(for $N = 2, 3, 4$ it is equal to 21, 78, 210, respectively)

vech and vec operators

- ⋆ vech is the operator that **stacks** the lower triangle of a $N \times N$ matrix as an $N(N+1)/2 \times 1$ vector:

$$\text{vech}H_t = (h_{11t}, h_{21t}, h_{22t}, h_{31t}, \ldots, h_{NNt})'$$

- ⋆ vec is the operator that **stacks** a matrix as column vector:

$$\text{vec}H_t = (h_{11t}, h_{21t}, \ldots, h_{N1t}, h_{12t}, h_{22t}, \ldots, h_{NNt})'$$

General VEC(1,1)

In this model, h_{ijt} is a **linear function** of the **lagged squared errors**, **cross products of errors**, and **lagged values of all the elements of** H_t.

The VEC(1, 1) is defined as:

$$\text{vech}H_t = c + A\,\text{vech}(\varepsilon_{t-1}\varepsilon'_{t-1}) + G\,\text{vech}H_{t-1},$$

where c is a $N^* \times 1$ vector of parameters [with $N^* = N(N+1)/2$], and A and G are $N^* \times N^*$ matrices of parameters.

Note: The VEC model is called VECH by some authors.

Stationarity conditions: VEC

- ⋆ The VEC(1,1) model

$$\text{vech}H_t = c + A\,\text{vech}(\varepsilon_{t-1}\varepsilon'_{t-1}) + G\,\text{vech}H_{t-1}$$

can be written as a VARMA(1,1) model for each $\text{vech}(\varepsilon_t\varepsilon'_t)$:

$$\text{vech}H_t = c + (A+G)\,\text{vech}(\varepsilon_{t-1}\varepsilon'_{t-1}) + w_t - G\,w_{t-1},$$

where $w_t = \text{vech}(\varepsilon_t\varepsilon'_t - H_t)$

- ⋆ The VEC(1,1) model is weakly stationary if the eigenvalue of $A+G$ are less than one in modulus. Then,

$$\text{vech}H = \mathsf{E}(\text{vech}H_t) = (I_{N^*} - A - G)^{-1}c$$

Positivity conditions for VEC (Gouriéroux, 1997)

- ⋆ After some **reparametrization**, a general matrix (rather than vech) expression of H_t in the VEC(1,1) case is:

$$H_t = C + (I_N \otimes \varepsilon'_{t-1})\tilde{A}(I_N \otimes \varepsilon_{t-1}) + \mathsf{E}_{t-2}\left[(I_N \otimes \varepsilon'_{t-1})\tilde{G}(I_N \otimes \varepsilon_{t-1})\right]$$

- ⋆ Hence, sufficient conditions for positivity of H_t are that C, $\tilde{A}$, and $\tilde{G}$ be **semi-positive-definite** (SPD), with at least one being **positive definite** (PD).

- ⋆ These restrictions may be imposed in estimation by using Cholesky factorizations. Usually **they are not imposed**, but they can be checked after the unrestricted estimation.

Diagonal and Scalar VEC

- ⋆ To reduce the number of parameters, Bollerslev, Engle, and Wooldridge (1988) suggest the diagonal VEC (DVEC) model in which the A and G matrices are **diagonal**.

- ⋆ Each **variance** h_{iit} depends only on its **own past squared error** $\varepsilon^2_{i,t-1}$ and its **own lag** $h_{ii,t-1}$. Each **covariance** h_{ijt} depends only on its **own past cross-products of errors** $\varepsilon_{i,t-1}\varepsilon_{j,t-1}$, and its **own lag** leading to no "spillover effects".

- ⋆ Big reduction: **9** parameters instead of **21** when $N = 2$; **18** instead of **78** when $N = 3, \ldots$

- ⋆ **Scalar** (full) VEC: $A = aU^*_N$ and $G = gU^*_N$ where a and g are **scalars** and U^*_N is a matrix of ones $(N^* \times N^*)$. (diagonal: $U^*_N = I^*_N$).

Diagonal bivariate VEC(1,1)

$$\begin{bmatrix} h_{11t} \\ h_{21t} \\ h_{22t} \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} + \begin{bmatrix} a_{11} & 0 & 0 \\ 0 & a_{22} & 0 \\ 0 & 0 & a_{33} \end{bmatrix} \begin{bmatrix} \varepsilon_{1,t-1}^2 \\ \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{2,t-1}^2 \end{bmatrix}$$
$$+ \begin{bmatrix} g_{11} & 0 & 0 \\ 0 & g_{22} & 0 \\ 0 & 0 & g_{33} \end{bmatrix} \begin{bmatrix} h_{11,t-1} \\ h_{21,t-1} \\ h_{22,t-1} \end{bmatrix}$$

or

$$h_{11t} = c_1 + a_{11}\varepsilon_{1,t-1}^2 + g_{11}h_{11,t-1}$$
$$h_{21t} = c_2 + a_{22}\varepsilon_{1,t-1}\varepsilon_{2,t-1} + g_{22}h_{21,t-1}$$
$$h_{22t} = c_3 + a_{33}\varepsilon_{2,t-1}^2 + g_{33}h_{22,t-1}$$

Diagonal bivariate VEC(1,1)

Equivalently,

$$\begin{pmatrix} h_{11t} & h_{21t} \\ h_{21t} & h_{22t} \end{pmatrix} = \begin{pmatrix} c_1 & c_2 \\ c_2 & c_3 \end{pmatrix} + \begin{pmatrix} a_{11} & a_{22} \\ a_{22} & a_{33} \end{pmatrix} \odot \begin{pmatrix} \varepsilon_{1,t-1}^2 & \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{1,t-1}\varepsilon_{2,t-1} & \varepsilon_{2,t-1}^2 \end{pmatrix}$$
$$+ \begin{pmatrix} g_{11} & g_{22} \\ g_{22} & g_{33} \end{pmatrix} \odot \begin{pmatrix} h_{11,t-1} & h_{21,t-1} \\ h_{21,t-1} & h_{22,t-1} \end{pmatrix}$$

In general,

$$H_t = C + A_d \odot (\varepsilon_{t-1}\varepsilon_{t-1}') + G_d \odot H_{T-1}$$

If one of C, A_d, and G_d (all of dimension $N \times N$) is PD, and the other two are SPD, H_t is PD.

Scalar DVEC: if $a_{ij} = a$ and $g_{ij} = g, \forall i,j$.

Bivariate BEKK(1,1,1) (Engle and Kroner, 1995)

$$
\begin{aligned}
\begin{pmatrix} h_{11t} & h_{21t} \\ h_{21t} & h_{22t} \end{pmatrix} &= \begin{pmatrix} \omega_{11} & \omega_{21} \\ \omega_{21} & \omega_{22} \end{pmatrix} \\
&+ \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}' \begin{pmatrix} \varepsilon_{1,t-1}^2 & \varepsilon_{1,t-1}\varepsilon_{2,t-1} \\ \varepsilon_{2,t-1}\varepsilon_{1,t-1} & \varepsilon_{2,t-1}^2 \end{pmatrix} \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix} \\
&+ \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}' \begin{pmatrix} h_{11,t-1} & h_{21,t-1} \\ h_{21,t-1} & h_{22,t-1} \end{pmatrix} \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}
\end{aligned}
$$

Note: 11 parameters, against 21 in the VEC model.

Bivariate BEKK(1,1,1)

Same linear structure as in VEC model...

$$
\begin{aligned}
h_{11t} = \omega_{11} &+ a_{11}^2 \varepsilon_{1,t-1}^2 + 2a_{11}a_{21}\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{21}^2 \varepsilon_{2,t-1}^2 \\
&+ g_{11}^2 h_{11,t-1} + 2g_{11}g_{21}h_{21,t-1} + g_{21}^2 h_{22,t-1} \\
h_{21t} = \omega_{21} &+ a_{11}a_{12}\varepsilon_{1,t-1}^2 + (a_{11}a_{22} + a_{12}a_{21})\varepsilon_{1,t-1}\varepsilon_{2,t-1} + a_{22}a_{21}\varepsilon_{2,t-1}^2 \\
&+ g_{11}g_{12}h_{11,t-1} + (g_{11}g_{22} + g_{21}g_{21})h_{21,t-1} + g_{22}g_{21}h_{22,t-1}
\end{aligned}
$$

... but constraints on parameters (compare with bivariate VEC).

General BEKK(1,1,K)

- ⋆ The BEKK(1,1,K) model is defined as:

$$H_t = \Omega + \sum_{k=1}^{K} A_k' \varepsilon_{t-1} \varepsilon_{t-1}' A_k + \sum_{k=1}^{K} G_k' H_{t-1} G_k$$

  where Ω is symmetric, A_k and G_k are $N \times N$ matrices of parameters.

- ⋆ H_t is PD if H_0 and Ω are PD. One can replace Ω by $C'C$, where C is lower triangular, to ensure that Ω is PD.

Remarks

- ⋆ Interpretation of the basic parameters **not direct**.
- ⋆ By increasing K, the model gets more flexible (e.g. for $N = K = 2$, there are 19 parameters, against 21 in the bivariate VEC).
- ⋆ Scalar (full) BEKK model: $A_k = a_k U_N, G_k = g_k U_N$
- ⋆ Diagonal BEKK model: A_k and G_k are diagonal matrices. It is a restrictive DVEC model.
- ⋆ Scalar diagonal BEKK: $A_k = a_k I_N$, $G_k = g_k I_N$. For $K = 1$,

$$H_t = \Omega + \alpha \varepsilon_{t-1} \varepsilon'_{t-1} + \gamma H_{t-1}$$

where $\alpha = a_1^2$ and $\gamma = g_1^2$, and $\alpha + \gamma < 1$ for stationarity.

Stationarity conditions: BEKK

- ⋆ The BEKK(1,1,1) model

$$H_t = \Omega + A'\varepsilon_{t-1}\varepsilon'_{t-1}A + G'H_{t-1}G$$

can be written as a VEC model (subject to restrictions) using the formula

$$\text{vec}(ABC) = (C' \otimes A)\text{vec}B,$$

so that

$$H_t = \Omega + (A \otimes A)'\text{vec}(\varepsilon_{t-1}\varepsilon'_{t-1}) + (G \otimes G)'\text{vec}\, H_{t-1}$$

- ⋆ Hence, the BEKK model is weakly **stationary** if the eigenvalues of $(A \otimes A) + (G \otimes G)$ are smaller than 1 in modulus, and then

$$\text{vec}H = \mathsf{E}(\text{vec}\, H_t) = (I_{N^2} - (A \otimes A)' + (G \otimes G)')^{-1}\text{vec}\, \Omega$$

Number of parameters

- ⋆ VEC(1,1): Dimensions $N = 2, 3, 4, 5$, parameters : 21, 78, 210, 465
- ⋆ BEKK(1,1,1): Dimensions $N = 2, 3, 4, 5$, parameters : 11, 24, 42, 65

Conditional correlations

- H_t can always be written as

$$\begin{aligned} H_t &= D_t R_t D_t \\ D_t &= \text{diag}(h_{11t}^{1/2}, \ldots, h_{NNt}^{1/2}) \text{ (a diagonal matrix)} \\ R_t &= (\rho_{ijt}), \text{ with } \rho_{iit} = 1 \end{aligned}$$

  R_t is the $N \times N$ matrix of **conditional correlations**, and h_{iit} is a conditional variance. Hence,

$$h_{ijt} = \rho_{ijt}\sqrt{h_{iit}h_{jjt}}, \quad \forall i \neq j$$

- Each h_{iit} can be defined **individually** as a GARCH equation.

- Positivity of H_t follows from positivity of R_t and of all h_{iit}.

Bivariate case

$$\begin{pmatrix} h_{11t} & h_{21t} \\ h_{21t} & h_{22t} \end{pmatrix} = \begin{pmatrix} \sqrt{h_{11t}} & 0 \\ 0 & \sqrt{h_{22t}} \end{pmatrix} \begin{pmatrix} 1 & \rho_{21t} \\ \rho_{21t} & 1 \end{pmatrix} \begin{pmatrix} \sqrt{h_{11t}} & 0 \\ 0 & \sqrt{h_{22t}} \end{pmatrix}$$

since $h_{21t} = \rho_{21t}\sqrt{h11t}\sqrt{h22t}$

Conditional variances

- ⋆ No spillover (or interaction) terms: h_{iit} a function of its **own lags** and the **lags of** ε_{it} only, for example a GARCH(1,1) equation. Like in DVEC and DBEKK.
- ⋆ Spillover terms: h_{iit} a function of the **lags of** h_{jjt} **and of** ε_{jt} both for $i = j$ and for $i \neq j$.
- ⋆ Spillover terms could also include **lags** of $\varepsilon_{it}\varepsilon_{jt}$ **and of** h_{ijt} for $i \neq j$, such as in the full VEC and BEKK models.
- ⋆ **Excluding** spillover terms renders the estimation of conditional correlation models easier but **prevents capturing interactions**.

CCC of Bollerslev (1990)

In this case,

$$R_t = R = (\rho_{ij}), \quad \rho_{ii} = 1,$$

i.e. Constant Conditional Correlations (CCC). Hence,

$$h_{ijt} = \rho_{ijt}\sqrt{h_{iit}h_{jjt}}, \quad \forall i \neq j,$$

and thus the **dynamics** of the **covariance** is determined only by the dynamics of the two conditional **variances**.

NB: there are $N(N-1)/2$ parameters in R.

(Scalar) DCC of Engle (2002)

DCC for Dynamic Conditional Correlations.

- ⋆ The DCC(1,1):
$$R_t = (\text{Diag}Q_t)^{-1/2} Q_t (\text{Diag}Q_t)^{-1/2}.$$
- ⋆ Note: $\text{Diag}Q_t = I_N \odot Q_t = \text{diag}(q_{11t}, \ldots, q_{NNt})$.
- ⋆ Q_t is a $N \times N$ matrix, symmetric and > 0, given by
$$Q_t = (1 - \theta_1 - \theta_2)\bar{Q} + \theta_1 u_{t-1} u'_{t-1} + \theta_2 Q_{t-1},$$
where $u_t = (u_{1t}, \ldots, u_{Nt})', u_{it} = \varepsilon_{it}/\sqrt{h_{iit}}$,
- ⋆ $\bar{Q}$ is a $N \times N$ matrix, **symmetric** and **PD**, of parameters, and θ_1 and θ_2 **positive parameters** satisfying $\theta_1 + \theta_2 < 1$, then Q_t and R_t are **PD**.

Remarks

- ⋆ Q_t **is not** a correlation matrix, since q_{iit} is not equal to 1 by construction. It is transformed into a correlation matrix
- ⋆ If $\theta_1 = \theta_2 = 0$, and $q_{ii} = 1$ for all i, the CCC model is obtained. Hence **one can test** for CCC against DCC. However, $\bar{q}_{ii} = 1$ for all i is not a necessary restriction for DCC.
- ⋆ All the correlations obey the same dynamics. This saves **a lot** of parameters, compared to VEC and BEKK models, but is restrictive (especially when N is large).

(Scalar) Asymmetric DCC (ADCC)

$$Q_t = \Omega + \theta_1 u_{t-1} u'_{t-1} + \gamma \eta_{t-1} \eta'_{t-1} + \theta_2 Q_{t-1},$$

where Ω is PD and symmetric, and $\eta_t = \min[u_t, 0]$.

If $\gamma > 0$, correlations increase more if both shocks are negative (declining market) than if they are positive.

Remark: In some models, it is difficult to introduce asymmetric effects.

Number of parameters

- ⋆ CCC: Dimensions $N = 2, 3, 4, 5$, parameters : 7, 12, 18, 25
- ⋆ DCC: Dimensions $N = 2, 3, 4, 5$, parameters : 9, 14, 20, 27

Other specifications

- ⋆ DCC of Tse and Tsui (2002)
- ⋆ cDCC of Aielli (2009)
- ⋆ GDCC of Capiello, Engle, and Sheppard (2007)
- ⋆ GDCC of Hafner an Franses (2009)
- ⋆ mDCC of Bauwens, Hafner and Pierret (2012)
- ⋆ One-factor DCC
- ⋆ ...

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Estimation of MGARCH models

- ⋆ **Maximum Likelihood Estimation** (MLE) requires an assumption about the density of z_t, denoted $g(z_t;\eta)$, where η is an additional parameter vector.

- ⋆ Maximize with respect to (θ,η) the function

$$L_T(\theta,\eta) = \sum_{t=1}^{T} \log f(y_t|\theta,\eta,I_{t-1}),$$

with

$$f(y_t|\theta,\eta,I_{t-1}) = |H_t|^{-1/2} g\left(H_t^{-1/2}(y_t-\mu_t);\eta\right)$$

where the dependence with respect to θ occurs through μ_t and H_t.

Gaussian likelihood

- ⋆ In many cases, $z_t \sim N(0, I_N)$ is assumed (hence η is empty). Then, neglecting a constant,

$$L_T(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left(\log|H_t| + (y_t - \mu_t)' H_t^{-1}(y_t - \mu_t)\right),$$

- ⋆ This Gaussian log-likelihood provides the **Quasi-ML** (QML) estimator that is consistent for θ even if the true density is not $N(0, I_N)$ (if μ_t and H_t are correctly specified).

- ⋆ However, this QML estimator is **less efficient** than the ML estimator that would be obtained using the log-likelihood based on the true density.

Remarks

- ⋆ For financial returns, normality is **not** realistic, like for univariate GARCH models.
- ⋆ For financial applications (such as computing the VaR), it is important to use the most correct assumption about the density. Hence, normality is **not** useful in some applications...
- ⋆ Alternative distributions: multivariate Student (to account for excess conditional kurtosis), multivariate skewed-Student or mixture of two multivariate Gaussian densities (for conditional skewness), generalized hyperbolic distribution, ...
- ⋆ **Danger** of this approach: if the assumption is not correct, **inconsistency** of the estimator results. To what extent?

Student density

- ⋆ The multivariate Student density, denoted $ST(0, I_N, \nu)$ (ν corresponds to η), is

$$g(z_t|\nu) = \frac{\Gamma\left(\frac{\nu+N}{2}\right)}{\Gamma(\frac{\nu}{2})\left[\pi(\nu-2)\right]^{\frac{N}{2}}}\left[1+\frac{z_t' z_t}{\nu-2}\right]^{-\frac{N+\nu}{2}}$$

  where $\Gamma(a) = \int_0^\infty x^{a-1}e^{-x}dx$ is the Gamma function.

- ⋆ Notice we impose $\nu > 2$, and $\text{var}(z_t) = I_N$ while $\text{E}(z_t) = 0$. Although uncorrelated, the elements of z_t are not independent.

- ⋆ When $\nu \to \infty$, $ST(0, I_N, \nu) \to N(0, I_N)$.

- ⋆ When $\nu \to 2$, the tails of the density become thicker and thicker.

Asymptotic properties of ML and QML

- ⋆ **Consistency** of QMLE is shown (Bollerslev and Wooldridge 1992; Jeantheau 1998).
- ⋆ **Asymptotic normality**: Comte and Lieberman (2003) for BEKK; Ling and McAleer (2003) for CCC with ARMA-GARCH; Francq and Zakoïan (2011) for CCC-AGARCH.
- ⋆ Asymptotic normality otherwise assumed in practice, hence inference done as usual (asymptotic χ^2 Wald and likelihood ratio tests).

Two step estimation of CCC/DCC models

- ⋆ This approach uses the Gaussian likelihood (ML under the normality assumption, or QML otherwise).
- ⋆ Substituting $H_t = D_t R_t D_t$ in the Gaussian log-likelihood function gives:

$$L_T(\theta) = -\frac{1}{2}\sum_{t=1}^{T}\left(\log|D_t R_t D_t| + u_t' R_t^{-1} u_t\right),$$

where $u_t = D_t^{-1}(y_t - \mu_t)$, so that $u_t' R_t^{-1} u_t = (y_t - \mu_t)' D_t^{-1} R_t^{-1} D_t^{-1}(y_t - \mu_t)$.

Two step estimation of CCC/DCC models

Hence, we can write:

$$
\begin{aligned}
L_T(\theta) &= -\frac{1}{2}\sum_{t=1}^{T}\left(\log|D_t R_t D_t| + u_t' R_t^{-1} u_t\right) \\
&= -\frac{1}{2}\sum_{t=1}^{T}\left(2\log|D_t| + u_t' u_t\right) \leftarrow Q1 L_T(\theta_V) \\
&= -\frac{1}{2}\sum_{t=1}^{T}\left(\log|R_t| + u_t' R_t^{-1} u_t - u_t' u_t\right) \leftarrow Q2 L_T(\theta_V, \theta_R)
\end{aligned}
$$

θ_V parameters of the conditional **variances** (in D_t),
θ_R parameters of the conditional **correlations** (in R_t).

Two step estimation of CCC/DCC models

- $L_T(\theta) = L_T(\theta_V, \theta_R) = Q1L_T(\theta_V) + Q2L_T(\theta_V, \theta_R)$
- **First** step: estimate θ_V by

$$\hat{\theta}_V = \text{argmax} Q1L_T(\theta_V)$$

  Easy: estimate separate univariate GARCH models if there is no spillover effects in conditional variances.

- **Second** step: estimate θ_R by

$$\hat{\theta}_R = \text{argmax} Q2L_T(\hat{\theta}_V, \theta_R)$$

  More easy than joint estimation, since many parameters are fixed in this step.

Remarks

- ⋆ **Consistency** and **asymptotic normality** of one step and two step estimators of DCC discussed in Engle (2002), Engle and Sheppard (2001) and Aielli (2009).
- ⋆ The two step estimators are **consistent** but **not efficient asymptotically**, since some information is sacrificed (about θ_V in the first step).
- ⋆ The variance matrix of the second step estimator of $\hat{\theta}_R$ has to be adjusted to take account of the first step, though this is not important for VaR point forecasts (but it is important for interval forecasts).

First step if no spillover in variance equations

- ⋆ Let $\theta_V = (\theta_{V1}, \theta_{V2}, \ldots, \theta_{VN})$ where θ_{Vi} contains the parameters of the GARCH equation for h_{iit}.

- ⋆ If there is no spillover term in h_{iit} for all i, the first step objective function can be written as the sum of N functions:

$$
\begin{aligned}
Q1L_T(\theta_V) &= -\frac{1}{2}\sum_{t=1}^{T}\left(2\log|D_t| + u_t' u_t\right) \\
&= -\frac{1}{2}\sum_{t=1}^{T}\sum_{i=1}^{N}\left(2\log h_{iit} + u_{it}^2\right)
\end{aligned}
$$

  Each function depends on different parameters (θ_{Vi}) and can be maximized separately.

Second step for the CCC model

- ⋆ In the CCC, θ_R is the constant (correlation) matrix R, and

$$Q2L_T(\hat{\theta}_V, R) = -\frac{1}{2}\sum_{t=1}^{T}\left(\log|R| + \hat{u}_t' R^{-1}\hat{u}_t\right),$$

  where $\hat{u}_t = \hat{D}_t^{-1}(y_t - \hat{\mu}_t)$ (**demeaned** and **degarched** data, obtained after step 1). $\hat{D}_t$ is D_t evaluated using $\hat{\theta}_V$

- ⋆ The maximizer of this, is the correlation matrix of the vectors $\hat{u}_t, t = 1, 2, ..., T$. Let $\tilde{R} = \frac{1}{T}\sum_{t=1}^{T}\hat{u}_t\hat{u}_t'$: it is not a correlation matrix (its diagonal elements are not equal to 1 for finite T).

- ⋆ The **estimate** of the **correlation matrix** is

$$\hat{R} = (\text{Diag}\tilde{R})^{-1/2}\tilde{R}(\text{Diag}\tilde{R})^{-1/2}.$$

One step or two step estimation?

- ⋆ Nothing prevents us in principle to estimate all the parameters of DCC models jointly in one step.
- ⋆ Estimation in one step gets difficult when N is large, because the number of parameters is
  - ⋆ $O(N^3)$ in the variance equations if all spillover terms are included, or $O(N^2)$ if only the squared errors and variances are included (i.e. the cross products and covariances are not included);
  - ⋆ $O(N^2)$ in the correlation model (due to the matrix $\bar{Q}$).
- ⋆ Even if all spillover terms are omitted in the variance equations, the number of parameters in step 2 is $O(N^2)$ and is a problem for large N. Targeting $\bar{Q}$ is a solution.

Variance targeting for a VEC model

⋆ In the VEC model, we know that

$$\text{vech}\, H = \mathsf{E}(\text{vech}\, H_t) = (I_{N^*} - A - G)^{-1} c$$

Hence we can write the VEC model as

$$h_t = (I_{N^*} - A - G)(\text{vech H}) + A\eta_{t-1} + Gh_{t-1}.$$

⋆ A consistent estimator of H, if $T > N$, is

$$\hat{H} = \frac{1}{T} \sum_{t=1}^{T} \hat{\varepsilon}_t \hat{\varepsilon}_t'$$

where $\hat{\varepsilon}_t = y_t - \hat{\mu}_t$, with $\hat{\mu}_t$ a consistent estimator of μ_t (easily obtained by OLS or QML).

Variance targeting for a scalar BEKK model

- ⋆ We can write the scalar BEKK model as

$$H_t = (1-\alpha-\beta)H + \alpha\varepsilon_{t-1}\varepsilon'_{t-1} + \gamma H_{t-1}$$

  since $\mathsf{E}(H_t) = H$ if $\alpha+\beta < 1$.

- ⋆ Whatever the value of N, this leaves only **two parameters** in the targeted QML objective function (i.e. the Gaussian quasi-likelihood where H is replaced by $\hat{H}$).

- ⋆ However, if N approaches T, $\hat{H}$ gets ill-conditioned (close to a singular matrix). Engle, Shephard and Sheppard (2008) "Fitting vast dimensional time-varying covariance models" (http://hdl.handle.net/2451/27851) show by simulations that this produces a strong downward bias in the targeted QMLE of α and β.

- ⋆ Note: Pedersen and Rahbek (2014) have developed the asymptotic theory for the BEKK model with variance targeting.

Outline

Introduction

Multivariate GARCH
- Motivation
- Definition
- Variance and covariance models
- Variance and correlation models

Estimation methods

Miscellany

Limits of MGARCH models

- ⋆ Curse of dimensionality:
  - ⋆ Engle and Sheppard (2001) estimate a scalar DCC model for 100 series by ML with targeting.
  - ⋆ Engle, Shephard and Sheppard (2008) estimate scalar BEKK and cDCC models for up to 480 series by the ML and Composite Likelihood methods (with targeting).
- ⋆ For $N > 10$: scalar (c)DCC/BEKK and factor models.
- ⋆ Note 1: scalar DVEC is the same as scalar DBEKK.
- ⋆ Note 2: scalar BEKK and DCC are quite similar (Caporin and McAleer, 2012)

Softwares for MGARCH models

- ⋆ R: “mgarchBEKK”, “rmgarch”, and “ccgarch”packages
- ⋆ Matlab: “MFE” Toolbox
- ⋆ OxMetrics: G@RCH
- ⋆ STATA
- ⋆ Some old softwares: RATS, GAUSS, EVIEWS

Development of MGARCH

- ⋆ More flexible DCC and factor models
- ⋆ Estimation methods for large dimensions
- ⋆ Copula-MGARCH models combine GARCH for variances and a (dynamic) copula for conditional dependence.
- ⋆ Finite mixtures of GARCH models.
- ⋆ Semi-parametric estimation.
- ⋆ Using high frequency data, typically intraday returns in daily MGARCH models (MIDAS, HEAVY).
- ⋆ Generalized Autoregressive Score (Dynamics Conditional Score) models: `gasmodel.com`

References

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- ⋆ Silvennoinen, A., and T. Teräsvirta (2009): "Multivariate GARCH Models". In Handbook of Financial Time Series, Ed. by T. G. Andersen, R. A. Davis, J.-P. Kreiss, and T. Mikosch. New York: Springer, 201-229.